

SERIES

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Lecture Outline

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Introduction

Definition (Series)

A series is a sum of terms of a sequence. That is given a sequence $\{U_n\}_{n \in \mathbb{N}}$, a series is the expression

$$\sum_{n=1}^{\infty} U_n = U_1 + U_2 + U_3 + \cdots$$

- 1 A series is **finite** if it has a finite number of terms otherwise **infinite**.
- 2 The sum of the first n terms is called the n th **partial** sum denoted by S_n .



Introduction

Let $\{U_n\}$ be an infinite sequence. Then the partial sum $S_1, S_2, \dots, S_n$ are defined as follows:

$$S_1 = U_1,$$

$$S_2 = U_1 + U_2,$$

$$\vdots$$

$$S_n = U_1 + U_2 + \dots + U_n = \sum_{k=1}^n U_k.$$

☛ An **infinite series** of general term U_n is denoted and defined by

$$\sum U_n = U_1 + U_2 + \dots + U_n + \dots = \sum_{k=1}^{\infty} U_k = \lim_{n \rightarrow \infty} S_n$$



Sum of an Arithmetic Sequence

The sum of the first n terms of an arithmetic sequence is given by

$$S_n = \frac{n}{2}(2U_1 + (n-1)d) \quad (1)$$

Sum of Geometric Sequence (Finite Sum)

The sum of the first n terms of a geometric sequence is given by

$$S_n = \frac{U_1(1-r^n)}{1-r}, \quad \text{for } r < 1 \quad (2)$$

$$S_n = \frac{U_1(r^n-1)}{r-1}, \quad \text{for } r > 1 \quad (3)$$



Example

The sequence

$$2, 5, 8, 11, \dots \quad (4)$$

is an arithmetic sequence with $U_1 = 2$ and $d = 3$.

The sum of the first 10 terms is:

$$S_n = \frac{n}{2}(2U_1 + (n-1)d) \quad (5)$$

$$S_{10} = \frac{10}{2}(2(2) + (9)3) \quad (6)$$

$$= 5(31) \quad (7)$$

$$= 155 \quad (8)$$



Telescoping series

Definition

A **telescoping series** is an infinite series of general term

$$u_n = a_n - a_{n+1} \quad (9)$$

where $\{a_n\}$ is a sequence.

We notice that the sum

$$s_n = u_1 + u_2 + u_3 + \cdots u_{n-1} + u_n \quad (10)$$

$$= (a_1 - a_2) + (a_2 - a_3) + (a_3 - a_4) \cdots + (a_{n-1} - a_n) + (a_n - a_{n+1}) \quad (11)$$

$$= a_1 - \overset{0}{\cancel{a_2}} + \overset{0}{\cancel{a_2}} - \overset{0}{\cancel{a_3}} + \overset{0}{\cancel{a_3}} - \overset{0}{\cancel{a_4}} + \cdots - \overset{0}{\cancel{a_{n-1}}} + \overset{0}{\cancel{a_{n-1}}} - \overset{0}{\cancel{a_n}} + a_n - a_{n+1} \quad (12)$$

$$= a_1 - a_{n+1} \quad (13)$$



Convergence of Geometric Series

- ① The series of general term

$$U_n = U_1 r^{n-1}, \quad n \geq 1, \quad (14)$$

is called **geometric series**.

- ② U_1 is the **first term** of the series and r is the **ratio**.

Theorem (Infinite Sum)

- ① Suppose $U_1 \neq 0$.
- ② The geometric series $\sum_{n=1}^{\infty} U_n$ converges if $|r| < 1$ (ie $-1 < r < 1$), and
- ③ diverges if $|r| \geq 1$.
- ④ and has sum $\frac{U_1}{1-r}$



Example

Show that the series of general term $u_n = 2e^{-3n}$ converges and hence find the sum.

Since is this a geometric series, we can use the ratio to determine convergence. The ratio is

$$r = \frac{u_{n+1}}{u_n} \quad (15)$$

$$= \frac{2e^{-3(n+1)}}{2e^{-3n}} \quad (16)$$

$$= e^{-3} < 1 \quad (17)$$

The sum is

$$S = \frac{2e^{-3}}{1 - e^{-3}} \quad (18)$$



Convergent Series (General Case)

Convergent series

- ① An infinite series with partial sums $S_1, S_2, \dots$ is **convergent** if

$$\lim_{n \rightarrow \infty} S_n = S \quad (19)$$

for some $S \in \mathbb{R}$.

- ② The series is **divergent** if the limit does not exist.
- ③ If a series converges and $\lim_{n \rightarrow \infty} S_n = S$, then S is called the **sum of the series**.



Example

Determine if the series $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n \cdot (n+1)} + \cdots$ converges.

The n th term decomposes into

$$U_n = \frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$$

Hence

$$\begin{aligned} S_n &= U_1 + U_2 + U_3 + \cdots + U_n \\ &= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n-1} - \frac{1}{n}\right) + \left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= 1 - \frac{1}{n+1} = \frac{n}{n+1} \end{aligned}$$

Therefore

$$S = \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{n}{n+1} = 1$$

and so the series converges.



Example

Find the sequence of partial sums and the sum of the series of general term

$$u_n = \frac{2}{4n^2 - 1} \text{ for } n \geq 1.$$

- ① Using partial fractions, the sequence $\{u_n\}$ becomes

$$u_n = \frac{1}{2n-1} - \frac{1}{2n+1} \quad (20)$$

Therefore,

$$s_n = \left(1 - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \cdots + \left(\frac{1}{2n-1} - \frac{1}{2n+1}\right) \quad (21)$$

$$= 1 - \frac{1}{2n+1} = \frac{2n}{2n+1} \quad (22)$$

- ② The sum is $\lim_{n \rightarrow \infty} s_n = 1$



Theorem

If the series $\sum u_n$ converges, then the sequence $\{u_n\}$ converges and $\lim_{n \rightarrow \infty} u_n = 0$.

Proof

$$s_n = (u_1 + u_2 + \cdots u_{n-1}) + u_n \quad (23)$$

$$s_n = s_{n-1} + u_n \quad (24)$$

$$\implies u_n = s_n - s_{n-1} \quad (25)$$

$$\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} s_n - \lim_{n \rightarrow \infty} s_{n-1} \quad (26)$$

Since

$$\lim_{n \rightarrow \infty} s_{n-1} = \lim_{n \rightarrow \infty} s_n = S \quad (27)$$

$$\implies \lim_{n \rightarrow \infty} u_n = S - S = 0 \quad (28)$$



Theorem

If $\lim_{n \rightarrow \infty} u_n \neq 0$ or *fails to exist*, then the series $\sum u_n$ *diverges*.

Example

The series of general terms $u_n = \frac{n+1}{n}$, $v_n = \left(1 + \frac{2}{n}\right)^n$, and $w_n = n^2 \sin n$ diverge since none of the three sequences has limit zero.

① $\lim_{n \rightarrow \infty} u_n = 1,$

② $\lim_{n \rightarrow \infty} v_n = e^2,$

③ $\lim_{n \rightarrow \infty} w_n$ does not exist



Theorem

If the series $\sum_{n=1}^{\infty} U_n$ and $\sum_{n=1}^{\infty} V_n$ converges with sums A and B respectively, then

- ① $\sum_{n=1}^{\infty} U_n \pm V_n$ converge with sum $A \pm B$
- ② $\sum_{n=1}^{\infty} cU_n$ converge with sum cA for $c \in \mathbb{R}$.

Do we always have to compute n th sums to examine convergence?
Now discuss how to determine convergence without computing sums.



Integral Test

We employ our knowledge of integration to determine if a series converge.

Theorem

If $f(x)$ is a positive valued, continuous and a decreasing function for $x \geq 1$, then the series

$$\sum_{x=1}^{\infty} f(x)$$

- ① converges if $\int_1^{\infty} f(x) dx$ converges
- ② diverges if $\int_1^{\infty} f(x) dx$ diverges

Example

Determine if the harmonic series $\sum_{x=1}^{\infty} \frac{1}{x}$ converges or diverges.

$$\int_1^{\infty} f(x) dx = \int_1^{\infty} \frac{1}{x} dx \quad (29)$$

$$= \lim_{b \rightarrow \infty} \int_1^b \frac{1}{x} dx \quad (30)$$

$$= \lim_{b \rightarrow \infty} \ln(x) \Big|_1^b \quad (31)$$

$$= \lim_{b \rightarrow \infty} \ln(b) \quad (32)$$

$$= \infty \quad (33)$$

Hence the series is diverges.



P-series

Theorem

The series

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = 1 + \frac{1}{2^p} + \frac{1}{3^p} + \cdots + \frac{1}{n^p} + \cdots$$

converges if $p > 1$ or diverges if $p \leq 1$.

Example

- ① $1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} + \cdots$ **converges**, because this is a p-series with $p > 1$
- ② $5 + \frac{5}{\sqrt{2}} + \frac{5}{\sqrt{3}} + \cdots + \frac{5}{\sqrt{n}} + \cdots$ **diverges**, because this series can be simplified as a p-series with $p < 1$



Basic comparison test (Direct comparison test)

This allows the use of known convergence of series to establish convergence of others.

Theorem

Let $0 < u_n \leq v_n$ for all integers $n \geq N$, where $N \in \mathbb{N}$.

① $\sum_N^{\infty} u_n \leq \sum_N^{\infty} v_n.$

② If $\sum_N^{\infty} v_n$ converges then $\sum_N^{\infty} u_n$ converges.

③ If $\sum_N^{\infty} u_n$ diverges then $\sum_N^{\infty} v_n$ diverges.

Example

Determine if the following converges or diverges

① $\sum_{n=1}^{\infty} \frac{1}{2+5^n}$

① Let $u_n = \frac{1}{2+5^n}$ and $v_n = \frac{1}{5^n}$ for $n \geq 1$.

② Notice that $u_n < v_n$ since

$$\frac{1}{2+5^n} < \frac{1}{5^n}$$

③ But v_n is a geometric sequence with $r = \frac{1}{5} < 1$, hence converges.

④ Thus u_n converges.

Note

Comparing this $U_n = U_1 r^{n-1}$ to $U_n = 1/5^n$, then for $n=1$ $U_1 = 1$ and for $r = U_2/U_1$ $r = 1/5$

Example

Use the direct comparison test to show that the series below converges $\sum_{n=1}^{\infty} \frac{1}{2^n + \sqrt{n}}$.

① For all $n \geq 1$,

$$2^n + \sqrt{n} \geq 2^n \quad (34)$$

$$\frac{1}{2^n + \sqrt{n}} \leq \frac{1}{2^n} \quad (35)$$

② Since the geometric series $\sum \frac{1}{2^n}$ converges, if this implies

③ $\sum \frac{1}{2^n + \sqrt{n}}$ converges as well.



Limit comparison test

Theorem (Limit Comparison Test)

Let $u_n > 0$ and $v_n > 0$ for all $n \geq N$ (N an integer.)

- 1 If $\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = l > 0$, then $\sum u_n$ and $\sum v_n$ both converge or both diverge.
- 2 If $\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = 0$ and $\sum v_n$ converges, then $\sum u_n$ converges.



Example

Determine if $\sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n^2+1}}$ converges or diverges

Let $v_n = \frac{1}{\sqrt[3]{n^2}}$ and $u_n = \frac{1}{\sqrt[3]{n^2+1}}$. Then

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{u_n}{v_n} &= \lim_{n \rightarrow \infty} \frac{\frac{1}{\sqrt[3]{n^2+1}}}{\frac{1}{\sqrt[3]{n^2}}} \\ &= \lim_{n \rightarrow \infty} \sqrt[3]{\frac{n^2}{n^2+1}} \\ &= 1 > 0\end{aligned}$$

and so u_n and v_n both diverge or converge. Precisely diverge since u_n diverges.



Alternating Series

The convergence tests that we have looked at so far apply only to series with positive terms. Now we learn how to deal with series whose terms are not necessarily positive. Of particular importance are alternating series, whose terms alternate in sign.

Definition (Alternating Series)

An alternating series is a series whose terms are alternately positive and negative.

Example

$$① \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

$$② \sum_{n=1}^{\infty} \frac{(-1)^{n-1} 3}{n^2} = 3 - \frac{3}{2} + 1 - \frac{3}{4} + \frac{3}{5} - \frac{1}{2} + \dots$$



Alternating Series Test

The following test says that if the terms of an alternating series decrease toward 0 in absolute value, then the series converges

Alternating Series Test

If the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n-1} b_n = b_1 - b_2 + b_3 - b_4 + \cdots; \quad b_n > 0 \quad (36)$$

satisfies

① $b_{n+1} \leq b_n$ for all n

② $\lim_{n \rightarrow \infty} b_n = 0$

then the series is convergent.



Example

The alternating harmonic series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$ converges because it satisfies the following

$$\textcircled{1} \quad b_{n+1} \leq b_n \text{ because } \frac{1}{n+1} < \frac{1}{n}$$

$$\textcircled{2} \quad \lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

Example

The alternating series $\sum_{n=1}^{\infty} \frac{(-1)^n 3n}{4n-1}$ is divergent because the second condition is not satisfied, that is

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} \frac{3n}{4n-1} = \lim_{n \rightarrow \infty} \frac{3}{4-1/n} = \frac{3}{4} \neq 0 \quad (37)$$

Absolute convergence

Given any series $\sum a_n$, we can consider the corresponding series

$$\sum_{n=1}^{\infty} |a_n| = |a_1| + |a_2| + |a_3| + \dots \quad (38)$$

whose terms are the absolute values of the terms of the original series.

Example

① Given $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$

② Then the absolute case is: $\sum_{n=1}^{\infty} \left| \frac{(-1)^{n-1}}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$

The is called a harmonic series.



Absolute convergence

Definition

A series $\sum_{n=1}^{\infty} a_n$ is **absolute convergent** if the series $\sum_{n=1}^{\infty} |a_n|$ converges

Example

Determine if $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2}$ is absolute convergent

- 1 Notice that $\sum_{n=1}^{\infty} \left| \frac{(-1)^{n-1}}{n^2} \right| = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots$
- 2 which is a convergent p -series.
- 3 Thus the series is absolute convergent.



Example

Determine if $\sum_{n=1}^{\infty} \frac{(-1)^n e^{-1}}{n^2}$ is absolute convergent.

- 1 Let $u_n = \frac{(-1)^n e^{-1}}{n^2}$.
- 2 Then $|u_n| = \left| \frac{(-1)^n e^{-1}}{n^2} \right| = \frac{e^{-1}}{n^2} = e^{-1} \frac{1}{n^2}$.
- 3 Since the p-series $\sum \frac{1}{n^2}$ converges then the series $\sum |u_n|$ converges.
- 4 Thus, $\sum u_n$ is absolute convergent.



Definition (Conditional convergence)

A series $\sum a_n$ is called conditionally convergent if it is convergent but not absolutely convergent.

Theorem

If a series $\sum a_n$ is absolutely convergent, then it is convergent.

Example

By the alternating series test, the series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$ is **convergent**, but the corresponding series of absolute values

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^{n-1}}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$$

is **not absolutely convergent** because this is p series with $p = 1$, and is therefore divergent.

Test for absolute convergence

Theorem (The ratio test)

- 1 If $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = L < 1$, then the series $\sum u_n$ is absolute convergent.
- 2 If $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = L > 1$, then the series $\sum u_n$ is absolute divergent.
- 3 If $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = 1$, the test is inconclusive.



Example

Determine if $\sum_{n=1}^{\infty} (-1)^n \frac{n^3}{3^n}$ is absolute convergent.

Let $a_n = (-1)^n \frac{n^3}{3^n}$. Then

$$\begin{aligned}\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} (n+1)^3}{3^{n+1}} \cdot \frac{3^n}{(-1)^n n^3} \right| \\&= \lim_{n \rightarrow \infty} \left| \frac{(n+1)^3}{3n^3} \right| \\&= \frac{1}{3} \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right)^3 \\&= \frac{1}{3} < 1\end{aligned}$$

Therefore the given series is absolute convergent.



Test of Absolute convergence

The following test is convenient to apply when n_{th} powers occur.

Theorem (The n_{th} root test)

- ① If $\lim_{n \rightarrow \infty} \sqrt[n]{|u_n|} = L < 1$, then the series $\sum u_n$ is absolute convergent.
- ② If $\lim_{n \rightarrow \infty} \sqrt[n]{|u_n|} = L > 1$, then the series $\sum u_n$ is absolute divergent.
- ③ If $\lim_{n \rightarrow \infty} \sqrt[n]{|u_n|} = 1$, the test is inconclusive.



Example

Test the convergence of the series $\sum_{n=1}^{\infty} \left(\frac{2n+3}{3n+2} \right)^n$

$$u_n = \left(\frac{2n+3}{3n+2} \right)^n \Rightarrow$$

$$\sqrt[n]{|u_n|} = \frac{2n+3}{3n+2} \quad (39)$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{|u_n|} = \lim_{n \rightarrow \infty} \frac{2n+3}{3n+2} \quad (40)$$

$$= \frac{2 + ^3/n}{3 + ^2/n} \quad (41)$$

$$= \frac{2}{3} < 1 \quad (42)$$

Thus, the given series converges by the Root Test.



Exercise

- 1 Use a geometric series to represent the repeating decimal $5.\overline{23} = 5.232323\dots$, $1.\overline{414}$ and $1.24\overline{4123}$ as the ratio of two integers.
- 2 Which of the following series diverge.
 a) $\sum_1^\infty \frac{n(n+1)}{(n+2)(n+3)}$ b) $\sum_0^\infty \cos(\pi n)$ c) $\sum_1^\infty \cos \frac{1}{n}$
- 3 Find s_n and determine if the series converges or diverges.
 a) $\sum_1^\infty \ln \sqrt{\frac{n+1}{n}}$ b) $\sum_1^\infty \frac{2n+1}{n^2(n+1)^2}$ c) $\sum^\infty \left(\cos \frac{\pi}{n} + \sin \frac{\pi}{n} \right)$.
- 4 For each of the geometric series, find a , $r(x)$ and the sum $S(x)$ of the series. Use the inequality $|r(x)| < 1$ to find the values of x for which the series converges.
 a) $\sum_0^\infty (-1)^n x^n$ b) $\sum_0^\infty 3 \left(\frac{x-1}{2} \right)^n$ c) $\sum_0^\infty \frac{(-1)^n}{2} \left(\frac{1}{3+\sin x} \right)^n$.
- 5 Use the integral test to study the convergence and divergence of the following series.
 1) $\sum_{n=1}^\infty \frac{2}{3n+5}$, 2) $\sum_{n=1}^\infty n e^{-\frac{n}{2}}$ 3) $\sum_{n=1}^\infty \frac{1}{n \ln n}$ 4) $\sum_{n=1}^\infty \frac{2}{n \ln^2 n}$
 5) $\sum_{n=2}^\infty \frac{1}{5n+10\sqrt{n}}$ 6) $\sum_{n=2}^\infty \frac{\sqrt{n}}{\ln n}$ 7) $\sum_{n=1}^\infty \frac{1}{n(1+\ln^2 n)}$ 8) $\sum_{n=1}^\infty \frac{\arctan n}{n^2+1}$.

END OF LECTURE
THANK YOU

